Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. What is the primary goal of topic modeling? 1 / 1 point

- Summarizing specific details in documents
- Discovering abstract themes in documents
- Enhancing the readability of texts
- Classifying documents by authorship

Nice work

Topic modeling aims to identify and extract underlying thematic structures in a collection of documents. These thematic structures are referred to as topics.

2. In an attempt to select the optimal number of topics for their LDA model, a researcher is analyzing the perplexity scores. What trend should they expect to observe as they increase the number of topics? 1 / 1 point

- Decreasing perplexity
- Constant perplexity
- Increasing perplexity

Nice work

As more topics are added to the model, it generally becomes better at modeling the text, which is reflected by a decrease in perplexity. The researcher should look for a point where the decrease in perplexity diminishes to determine the optimal number of topics.

3. In the context of evaluating topic models, what advantage does measuring topic coherence offer over traditional statistical measures like perplexity? 1 / 1 point

- Assesses semantic relationships
- Enhances data preprocessing
- None of the above
- Simplifies model parameters

Nice work

Topic coherence measures assess the semantic relationships and relevance among the terms within a topic, providing a more qualitative evaluation of how meaningful and cohesive each topic is, which is closer to (although still different from) human evaluation criteria for topic quality.

4. What role does the multinomial distribution play in the topic models that are part of the Latent Dirichlet Allocation model? 1 / 1 point

- Determines word selection probability
- Calculates document similarity
- Assigns topics to documents
- Estimates topic coherence

Nice work

Within each topic model in Latent Dirichlet Allocation, the multinomial distribution is used to determine the probability of selecting each word from a topic, based on the frequency and distribution of words within that topic.

5. What is the role of the Dirichlet distribution in Latent Dirichlet Allocation? 1 / 1 point

- It calculates the similarity between different documents
- It classifies documents into predefined categories
- It models the distribution of topics in documents

Nice work

In LDA, the Dirichlet distribution is used to model the probabilities of different topics within documents, reflecting how topics are mixed in each document.

6. What is a challenge in using the output of topic modeling that can be difficult to solve in an automated way? 1 / 1 point

- Interpreting the topics
- Collecting sufficient data
- Calculating model accuracy

Nice work

A significant challenge in topic modeling is interpreting the topics generated by the model, as it often requires human judgment to understand and label the topics effectively.